

```

using System;
namespace SemesterTest
{
    public class Program
    {
        static void Main(string[] args)
        {
            Sales sales;

            Batch batch1;
            Batch batch2;
            Batch batch3;

            Transaction trans1;
            Transaction trans2;
            Transaction trans3;
            Transaction trans4;
            Transaction trans5;

            sales = new Sales();

            batch1 = new Batch("#0001", "Books of Science");
            batch2 = new Batch("#04000", "Books of Electronics");
            batch3 = new Batch("#3202", "More Books");

            trans1 = new Transaction("#123", "Physics", 15.70m);
            trans2 = new Transaction("#167", "Biology", 23.40m);
            trans3 = new Transaction("#014", "Glass Making", 143.25m);
            trans4 = new Transaction("#198", "Woodworking", 184.00m);
            trans5 = new Transaction("#018", "Mathematics", 19.00m);

            sales.Add(batch1);
            sales.Add(batch2);
            sales.Add(trans3);
            sales.Add(trans4);

            batch1.Add(trans1);
            batch1.Add(trans2);
            batch1.Add(batch3);
            batch3.Add(trans5);

            Console.WriteLine("Press key to Print Order");
            Console.ReadLine();
            sales.PrintOrders();
            Console.ReadLine();
        }
    }
}

```

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namespace SemesterTest

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{
    public class Sales
    {
        private List<Thing> _orders;

        public Sales()
        {
            _orders = new List<Thing>();
        }

        public void Add(Thing thing)
        {
            _orders.Add(thing);
        }

        public void PrintOrders()
        {
            Console.WriteLine("Sales:");
            decimal i = 0;
            foreach (Thing t in _orders)
            {
                i += t.Total();
                t.Print();
            }
            Console.WriteLine($"Sales total: ${i}");
        }
    }
}

```

```

namespace SemesterTest
{
    public abstract class Thing
    {
        private string _number;
        private string _name;

        public Thing(string number, string name)
        {
            _number = number;
            _name = name;
        }

        public abstract void Print();

        public abstract decimal Total();

        public string Number { get { return _number; } }
        public string Name { get { return _name; } }
    }
}

```

```
namespace SemesterTest
{
    public class Batch : Thing
    {
        private string _number;
        private string _name;
        private List<Thing> _items;

        public Batch(string number, string name) : base(number, name)
        {
            _number = number;
            _name = name;
            _items = new List<Thing>();
        }

        public void Add(Thing single)
        {
            _items.Add(single);
        }

        public override void Print()
        {
            Console.WriteLine($"Batch Sale: {Number}, {Name}");
            if (_items.Count == 0)
            {
                Console.WriteLine("Empty Order.");
            }
            else
            {
                int i = 0;
                foreach (Thing t in _items)
                {
                    i++;
                    if (t is Transaction)
                    {
                        Console.WriteLine($"#{i}, {t.Name}, ${t.Total()}");
                    }
                    else
                    {
                        Console.WriteLine($"#{i}, Batch: {t.Number}, {t.Name}, ${t.Total()}");
                    }
                }
                Console.WriteLine($"Total: ${Total()}");
            }
        }

        public override decimal Total()
        {
            decimal i = 0;
            foreach (Thing t in _items)
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        {
            i += t.Total();
        }
        return i;
    }

    public string Number { get { return _number; } }
    public string Name { get { return _name; } }
}
}

```

```

namespace SemesterTest
{
    public class Transaction : Thing
    {
        private string _number;
        private string _name;
        private decimal _amount;

        public Transaction(string number, string name, decimal amount) :
base(number, name)
        {
            _number = number;
            _name = name;
            _amount = amount;
        }

        public override void Print()
        {
            Console.WriteLine($"{Number}, {Name}, ${Total()}");
        }

        public override decimal Total() { return _amount; }

        public string Number { get { return _number; } }
        public string Name { get { return _name; } }
    }
}
}

```